

CONSORT AGDM, AB, Canada

WMO: 712230

Lat: 51.9369N

Lon: 110.7131W

Elev: 772

StdP: 92.39

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-30.1	-27.3	-33.9	0.2	-29.9	-31.0	0.2	-27.3	14.0	-5.7	12.2	-6.5	4.1	290	0.743

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	30.0	16.8	27.8	16.5	25.7	15.6	18.8	26.1	17.7	24.9	16.7	23.5	5.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.3	12.7	21.6	15.1	11.8	20.8	14.0	11.0	19.4	56.7	26.3	53.1	25.1	49.7	23.4	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 12.6	11.0	9.7		-34.1	33.4	3.1	2.3	-36.3	35.1	-38.1	36.4	-39.8	37.7	-42.0	39.4	
(5)			DB	-34.3	21.0	3.0	1.3	-36.4	21.9	-38.2	22.7	-39.9	23.4	-42.0	24.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-11.2	-11.3	-4.2	4.0	10.6	14.5	17.8	16.6	11.5	4.3	-4.9	-10.9
	DBStd	12.30	8.49	7.65	7.67	5.32	4.33	3.16	3.11	3.59	4.80	5.37	7.14	7.98
	HDD10.0	3253	657	596	441	188	46	3	0	1	37	187	448	649
	HDD18.3	5610	916	829	699	431	240	121	48	77	210	435	698	907
	CDD10.0	750	0	0	0	7	64	138	241	206	83	11	1	0
	CDD18.3	64	0	0	0	0	1	5	30	23	5	0	0	0
	CDH23.3	1074	0	0	1	2	49	98	429	359	130	6	0	0
	CDH26.7	311	0	0	0	0	6	18	137	109	42	0	0	0
(14) Wind	WSAvg	4.8	5.2	4.8	5.0	5.3	5.1	4.8	4.3	4.2	4.5	5.0	5.0	4.9
(15) Precipitation	PrecAvg	330	14	8	13	21	37	73	53	43	25	14	13	11
	PrecMax	471	47	18	36	61	80	150	128	84	83	33	44	18
	PrecMin	172	2	1	3	3	3	17	20	5	1	4	0	2
	PrecStd	75	10	5	7	15	23	32	24	21	21	7	11	5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.5	5.7	14.6	22.5	27.3	29.1	32.9	32.7	30.9	23.7	13.7	6.4
		MCWB	2.2	2.2	7.5	10.7	13.5	16.3	17.8	17.0	15.3	12.1	6.6	3.1
	2%	DB	3.7	3.1	10.9	19.3	24.6	26.0	30.1	29.5	27.2	18.9	9.0	4.0
		MCWB	1.3	0.7	5.1	9.1	12.3	15.6	18.0	16.3	14.2	10.1	4.3	1.6
	5%	DB	2.3	1.2	7.8	16.2	22.2	23.9	27.8	27.2	23.7	15.5	6.0	2.1
		MCWB	0.3	-0.6	3.6	7.3	11.5	14.8	17.4	16.1	13.2	8.4	2.5	-0.1
	10%	DB	0.5	-0.6	4.8	13.3	19.7	21.9	25.5	25.0	20.6	12.6	3.8	-0.3
		MCWB	-1.1	-2.0	1.9	5.9	10.3	13.7	16.7	15.5	11.9	6.8	1.1	-1.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	2.7	8.2	11.2	14.6	18.6	21.2	19.5	16.4	12.9	7.1	3.3
		MCDB	4.9	5.4	14.3	21.2	23.9	25.3	28.4	26.3	26.8	22.0	12.5	5.8
	2%	WB	1.5	0.9	5.4	9.4	13.3	16.9	19.3	18.2	14.9	10.7	4.7	1.7
		MCDB	3.5	2.9	10.2	18.4	22.3	23.6	26.5	25.8	25.0	17.4	8.8	3.9
	5%	WB	0.4	-0.4	3.6	7.8	12.2	15.6	18.3	17.1	13.6	8.9	2.7	-0.1
		MCDB	2.2	1.1	7.3	14.9	20.7	22.1	25.5	24.7	22.4	14.8	5.3	1.9
	10%	WB	-1.1	-2.1	2.0	6.3	11.0	14.5	17.3	16.1	12.4	7.3	1.3	-1.8
		MCDB	0.5	-0.7	4.8	12.4	18.7	20.5	24.0	23.3	19.9	12.4	3.5	-0.3
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	9.5	9.8	11.9	12.8	11.4	12.9	13.1	13.0	11.5	9.1	8.7
		MCDBR	10.0	10.8	13.5	17.3	16.8	15.0	16.8	17.5	18.6	16.5	12.1	10.4
	5% WB	MCWBR	8.3	8.9	8.4	8.7	7.6	7.0	6.9	7.0	8.2	8.9	8.1	8.1
		MCDWR	10.2	10.6	12.6	15.9	14.8	13.3	14.5	15.0	17.1	15.8	11.3	10.5
(40) Clear-Sky Solar Irradiance	taub	taub	0.257	0.304	0.291	0.340	0.366	0.370	0.362	0.377	0.322	0.284	0.259	0.250
		taud	2.341	2.225	2.439	2.326	2.301	2.335	2.396	2.321	2.454	2.538	2.469	2.340
	Ebn at Noon	Ebn at Noon	768	806	915	897	886	882	884	848	866	846	773	712
		Edh at Noon	59	90	89	113	122	119	111	113	88	66	52	50
(44) All-Sky Solar Radiation	RadAvg	1.14	2.20	3.61	4.81	5.82	6.02	6.42	5.22	3.73	2.25	1.17	0.86	
	RadStd	0.10	0.18	0.27	0.42	0.39	0.39	0.29	0.40	0.36	0.29	0.11	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page